MEDSURE — Hackathon Pitch & Presentation Playbook

STAGE GUIDE

Know the Fair Price. Verify Every Bill. | ■■■ ■■■ ■■■■■ ■■ ■■■ ■■■■■■

2-Min Pitch + Slide Deck + Q&A Defense

1. The 30-Second Elevator Pitch (Quick Round Hook)

Speaker Delivery:
"Judges, imagine your family member gets discharged from the hospital, and you're handed an itemized bill for ***■3,40,000****. You suspect you're being overcharged, but medical bills are full of cryptic medical codes, bundled consumable kits, and subtle arithmetic errors.* ***Over 70% of healthcare in India is paid out-of-pocket****, pushing 55 million citizens into debt each year.*

Meet ***MedSure*** *— India's first fair-price medical bill audit platform. With MedSure, patients know the fair benchmark price before admission, and audit every single hospital charge after discharge using a* ***100% deterministic, zero-hallucination audit engine****. We turn vulnerable patients into empowered, informed advocates."*

2. The 2-Minute Stage Pitch Script (Timed Delivery Breakdown)

Timeline	Stage Phase	Speaker Script & Visual Action
0:00 – 0:25 (25 sec)	**The Hook & Problem** *The Medical Blindspot*	"Healthcare is the only industry where you consume a service without knowing the price, and you pay a bill you cannot understand. In India, medical inflation exceeds 14%, and sudden hospital bills drive millions into debt. Even insured families face arbitrary claim rejections and unbundled consumables." *Action: Display confusing 10-page medical invoice on slide.*
0:25 – 0:45 (20 sec)	**The Solution** *Two-Act Architecture*	"We built **MedSure** with a singular mission: **'Know the Fair Price. Verify Every Bill. (■■■ ■■■ ■■■■■ ■■ ■■■ ■■■■■■■)'**. MedSure operates across two distinct acts: **Act 1: Before Admission** — Bilingual Clinical Guidance & Fair Price Bounds. **Act 2: After Discharge** — Deterministic Financial Bill Audit & Dispute Resolution."
0:45 – 1:15 (30 sec)	**Live Demo: Act 1** *Safety & Tariff Bounds*	"A patient enters symptoms in Hindi or English (e.g. '■■■■■ ■■■ ■■■ ■■■■'. **1. Clinical Safety Guard:** Scans for emergency red flags (chest pain, breathing difficulty). If found, price estimation is halted immediately to direct the patient to 112 emergency care. **2. Procedural Bounds:** For elective care, maps symptoms to Total Knee Replacement with official CGHS/PM-JAY reference bounds (■1.25L – ■1.85L). Zero guessing, zero hallucination."
1:15 – 1:45 (30 sec)	**Live Demo: Act 2** *Deterministic Audit Engine*	"After discharge, the patient uploads the bill. Our PyMuPDF pipeline itemizes charges, and our pure-Python Decimal engine applies **Rules A to E**: • **Rule A (Arithmetic):** Catches surgeon fee of 1 × ■70k billed as ■85k (+■15k error). • **Rule B (Duplicate):** Flags standalone Operation Theatre fee charged twice in the same session (+■35k error). • **Rule C (Tariff Breaches):** Flags implant price exceeding NPPA/CGHS statutory price caps (+■30k error). • **Rule D (Unbundling):** Flags surgical drapes billed on top of all-inclusive package surgery. Total verified discrepancy detected: **■92,500.00!**"
1:45 – 2:00 (15 sec)	**Closing & Action** *Empowering the Patient*	"We don't just find errors; we give the patient tools to act. MedSure generates a **signed ReportLab PDF Audit Report** and an editable **Hospital Dispute Letter** ready for the billing desk. MedSure transforms billing asymmetry into patient empowerment. Thank you!"

3. 5-Slide Presentation Deck Architecture

Slide #	Slide Title	Key Visuals & Core Talking Points
Slide 1	**The Medical Billing Blindspot**	• 70% Out-of-pocket healthcare expenditure in India; 55M pushed into poverty. • Information asymmetry: Patients cannot understand cryptic medical bills.
Slide 2	**MedSure: The Two-Act Solution**	• Act 1 (Before Admission): Fair Price Estimator + Emergency Red-Flag Triage. • Act 2 (After Discharge): Deterministic Financial Audit + Dispute Letter Resolution.

Slide 3	**Act 1: Bilingual Triage & Price Bounds**	• Hindi & English symptom intake with emergency red-flag safety screening. • Official reference datasets: CGHS, PM-JAY, and Standardized Benchmark tariffs.
Slide 4	**Act 2: Deterministic Audit (Rules A–E)**	• **Zero LLM Hallucinations:** Decimal-exact Python financial engine. • Detects arithmetic errors, double-billed OT charges, rate cap breaches, unbundling. • Case study: ■92,500 overcharge identified on standard synthetic Knee Replacement bill.
Slide 5	**Actionable Impact & Business Model**	• Instant signed PDF report + formal editable dispute letter to hospital billing. • **B2C:** Freemium patient advocate tool; **B2B:** API for Insurance TPAs & Corporate Health.

4. Judge Defense Matrix (Toughest Questions & Winning Responses)

Judge Question	Winning Answer Strategy & Direct Response
Q1: Why not just use ChatGPT or Claude to audit the bill?	**Strategy: Emphasize Financial Determinism vs. LLM Hallucination.** “In financial and medical matters, a single wrong number destroys trust. LLMs are non-deterministic and notoriously hallucinate arithmetic (multiplication of float decimals). MedSure uses a **100% deterministic Python Decimal engine** with statutory tariff tables (CGHS, NPPA price caps). We never guess or hallucinate rates.”
Q2: Will private hospitals accept your audit report?	**Strategy: Position as Respectful Inquiries, Not Accusations.** “MedSure does not accuse hospitals of fraud. It generates an **itemized clarification request** citing line numbers, multiplication discrepancies, and duplicate entries. In practice, billing departments routinely issue revised invoices once presented with mathematical proof rather than vague complaints.”
Q3: What is the commercialization / business model?	**Strategy: Highlight High-Margin B2B Insurance & B2C Freemium.** “1. **B2C:** Freemium model for patients (first bill audit free, nominal ■99–■199 for dispute filing package). 2. **B2B (Enterprise):** Health insurance TPAs (Third-Party Administrators) and corporate employers use our API to automate pre-auth invoice auditing, catching millions in duplicate and unbundled charges.”
Q4: What if a patient in a medical emergency checks prices?	**Strategy: Safety-First Clinical Ethics.** “Clinical safety precedes financial transparency. Our bilingual Triage Engine screens for red-flag keywords (chest pain, stroke signs, difficulty breathing). In emergency scenarios, price estimation is intentionally disabled, and prominent emergency care guidance (112 / 102) is displayed immediately.”